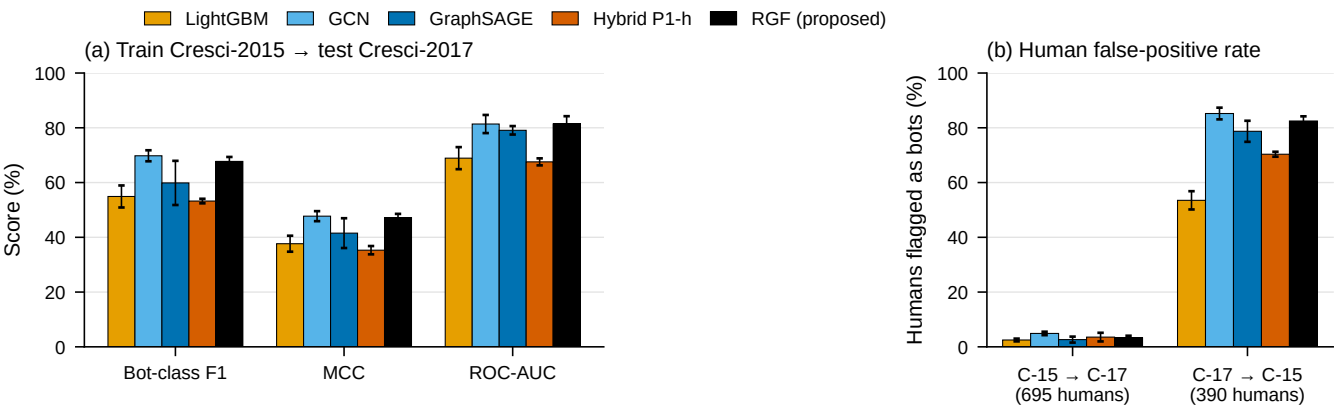
LightGBM
GCN
GraphSAGE
Hybrid P1-h
RGF (proposed)
(a) Train Cresci-2015 → test Cresci-2017
Score (%)
100
80
60
40
20
0
Bot-class F1
MCC
ROC-AUC
(b) Human false-positive rate
Humans flagged as bots (%)
100
80
60
40
20
0
C-15 → C-17
(695 humans)
C-17 → C-15
(390 humans)